CLASSROOM PLANETARY CLOCK & INTERVAL SYSTEM

Kinesthetic Time, Epicyclic 12:1 Gears & Modular Arithmetic

CARD 03 / 10 • TIME-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · **3·Clock** · 4·Fractions · 5·Primes · 6·Algebra · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

EPICYCLIC 12:1 GEARS & MODULO ARITHMETIC

Time is continuous rotational motion & modular math.
StackCalc Clock features an epicyclic gearset:
12:1 Exact Planetary Reduction Ratio
When minute hand turns 360°, hour hand advances 30°!

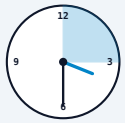
CALIBRATED 60-MINUTE / 12-HOUR DATUM

- Minute Unit Datum: 1 Minute = 6° dial sector
- Hour Mark: 1 Hour Mark = 30° arc (12 total sectors)
- Quadrant: 1/4 Hour Wedge = 90° right-angle sector
- Gear Ratio: 12.000:1 compound planetary gearset

CLOCK STAND LID & MODULAR WEDGE BAYS

Storage lid features easel stand & wedge bays.
Drop wedges into dial to measure elapsed time!

PLANETARY GEARS: 12:1 RATIO AT 3:30 (105°)



12:1 EPICYCLIC GEARS

360° min → 30° hour
At 3:30: Hour hand = 105°

15-Min Quadrant Wedge:

90° arc = 1/4 of an hour
Modulo 12 & Modulo 60

2. KIT PARTS INVENTORY (18 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Geared Clock Frame	1x	105.0 mm Dial (12h/60m)	Slate Gray
12:1 Epicyclic Gear	1x	Internal Sun-Planet Mesh	Gold / Blue
15-Min Wedges	4x	90° Quadrant (1/4 Hour)	Sky Blue
30-Min Wedges	2x	180° Semicircle (1/2 Hour)	Royal Purple
5-Min Wedges	8x	30° Hour Sector (1/12th)	Emerald Green
Display Stand Lid	1x	Clock Stand & Wedge Bay	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 16 Clock & Wedge Components + Storage Bin + Stand Lid = 18 Parts

FRACTION-ALIGNED TIME PALETTE

- Blue Wedges: 15-minute quadrants (90° right angles).
- Purple Wedges: 5-minute segments (30° hour steps).
- Emerald Face: Calibrated clock dial with 6° ticks.
- High-Visibility Hands: White minute & orange hour hands.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Seat the clock frame into its recessed bottom cradle. Drop the 14 colored wedges into their labeled radial lid bays. Any missing wedge leaves an empty bay. Easel lid latches firmly for silent, rattle-free carry.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

MODULO ARITHMETIC & CONTINUOUS RATIOS

Clock arithmetic is modulo 12 and modulo 60 governed by strict continuous gear kinematics.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 03 OF 10 • DOMAIN: KINEMATIC GEAR RATIOS & MODULAR ARITHMETIC • 12:1 EPICYCLIC REDUCTION • 15-MIN QUADRANT WEDGES

LESSON GUIDE: ELAPSED TIME & PLANETARY GEAR RATIOS

Step-by-Step Hands-On Tutorial: 3:30 + 0:45 = 4:15 & Modular Time

Grades 2–6 Elapsed Time & Modular Math • Hands-On Classroom & Home STEM Studio Series

CARD 03 / 10 • TIME-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/classroom-clock/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.3.MD.A.1 (Elapsed Time)
- CCSS.MATH.CONTENT.4.MD.A.2 (Time Intervals)
- TEKS 111.5.b.7.C (Elapsed Time Solutions)

Grades 2–6 Elapsed Time & Modular Math

MATHEMATICAL INQUIRY

How far does the hour hand travel when the minute hand turns one full circle around the dial?

1 STEP 1: SET START TIME TO 3:30

Rotate the minute hand to 6 (bottom of dial, 30 minutes).
Observe the internal gears: the hour hand advances smoothly halfway between 3 and 4 (exactly 105° on the dial).

2 STEP 2: INSERT 15-MINUTE WEDGES

We need to add 45 minutes of elapsed time.
Insert three 15-minute wedges (3 × 15 min = 45 min).
Wedges cover 3 × 90° = 270° angular arc around dial.
Notice: The wedges bridge past the 12 o'clock mark!

3–4 STEPS 3 & 4: ADVANCE GEARS & READ RESULT

3. ADVANCE MINUTE HAND: Turn hand clockwise across the 3 wedges past 4:00 to 4:15!
4. READ FINAL TIME: Hour hand at 4:15; minute hand at 3 (15 min).
Calculator Verification: Press Clock HMS Icon ([🕒] / [g][▸HMS] / [f][▸HR]) and run 3.30 [f] [▸HR] 0.45 [f] [▸HR] [+] [g] [▸HMS] → 4.15.

5 CORE PRINCIPLE: TIME IS CONTINUOUS KINEMATIC RATIO

Digital clocks make students think hour hands jump instantly when minutes reach 60. The planetary gears physically prove that time is a continuous ratio! Because 45 minutes is 3/4 of an hour, the hour hand moves through exactly 3/4 of its 30° sector (22.5°), landing precisely at 4:15!

★ EXTENSION INVESTIGATION: ELAPSED TIME: 8:45 + 1:30

Set clock to 8:45. Add two 45-minute wedges (90 min). Where does the planetary gear place the hour hand? (Scan QR for 3D animated solution!)

→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

stackcalc.io/parts

Classroom Lab Crates: stackcalc.io/schools